

Nuclear Burning Stages of Stars: pp Chain, CNO Cycles, Shell Model and Gamma Decay

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July 3, 2026

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Outline

Recap and Temperature Dependence of $p + p$

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The CN Cycle

CNO Bi-cycle and Additional Cycles

NeNa and MgAl Cycles

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Gamma Decay

Summary

Recap and Temperature Dependence of $p + p$

Recap: Where We Stand

From Lecture 22 we established:

- Net process $4p \rightarrow {}^4\text{He}$, $Q = 26.73$ MeV; effective $Q_{\text{eff}} = 26.2$ MeV after losing $2\langle E_\nu \rangle$ to escaping neutrinos
- First step $p + p \rightarrow d + e^+ + \nu$ proceeds via the weak interaction with $Q = 1.44$ MeV
- $\tau_{pp} \approx 10^{10}$ y at solar interior conditions – the chain's rate-limiting step
- Sun's hydrogen-burning lifetime $\approx 10^{10}$ y (10% core hydrogen $\Rightarrow$ middle-aged star)

In this set (Lectures 23–25) we now ask:

- Once d and ${}^3\text{He}$ are produced, how are they destroyed?
- What sets the relative importance of the pp-I, pp-II, and pp-III branches?
- How do CNO cycles (and their extensions) synthesize elements up to silicon?
- What tools – shell model, gamma-decay selection rules – do we need to describe the transitions involved?

Temperature Dependence of the $p + p$ Reaction

For a non-resonant charged-particle reaction:

$$\varepsilon \propto \langle \sigma v \rangle_{pp} \propto T^{-2/3} e^{-\tau}, \quad \tau = \frac{3E_0}{kT} = 42.46 (Z_1^2 Z_2^2 \mu / T_6)^{1/3}$$

Plotting ε vs. T_6 for $p + p$ vs. the CNO cycle reveals a **crossover at $T_6 \approx 20$** :

- $T_6 \lesssim 20$: $p + p$ dominates energy production (Sun and lower-mass stars, $M \lesssim M_\odot$)
- $T_6 \gtrsim 20$: CNO cycle takes over (more massive main-sequence stars)

Five interlocked quantities along the main sequence:

$$T \uparrow \Leftrightarrow M \uparrow \Rightarrow \text{fuel consumption rate} \uparrow, \text{ energy production} \uparrow, \text{ lifetime} \downarrow$$

More massive stars consume their fuel faster $\Rightarrow$ shorter lifetimes – traceable directly through the HR diagram.

Deuterium Burning

Destruction Channels for Deuterium

Deuterium is *produced* by $p + p \rightarrow d + e^+ + \nu$. The available destruction channels (all exothermic):

Reaction	Q -value (MeV)
$p + d \rightarrow {}^3\text{He} + \gamma$	5.5
$d + d \rightarrow {}^4\text{He} + \gamma$	23.8
$d + d \rightarrow p + t$	4.03
$d + d \rightarrow {}^3\text{He} + n$	3.27
${}^3\text{He} + d \rightarrow {}^4\text{He} + p$	18.35
${}^3\text{He} + d \rightarrow {}^5\text{Li} + \gamma$	16.38
${}^4\text{He} + d \rightarrow {}^6\text{Li} + p$	1.47

Because **protons are by far the most abundant** reactant, the dominant destruction channel is



Quasi-Equilibrium of the D/H Ratio

Time rate of change of deuterium:

$$\frac{dD}{dt} = r_{pp} - r_{pd} = \frac{H^2}{2} \langle \sigma v \rangle_{pp} - HD \langle \sigma v \rangle_{pd}$$

This is a **self-regulating system** reaching quasi-equilibrium ($dD/dt = 0$):

$$\boxed{\left(\frac{D}{H}\right)_e = \frac{\langle \sigma v \rangle_{pp}}{2 \langle \sigma v \rangle_{pd}}}$$

Interpretation of the strength hierarchy:

- $p + p$ is a weak-interaction reaction $\Rightarrow$ slow production
- $p + d$ is an EM-interaction reaction $\Rightarrow$ fast destruction
- Numerically: $(D/H)_e \approx 10^{-18}$ at $T_6 = 5$, i.e. $\ll 1$

Lifetimes reflect the same hierarchy:

$$\tau_H(\text{against } p) \approx 10^{10} \text{ y}, \quad \tau_d(\text{against } p) \approx 1.6 \text{ s}$$

The Observed D/H Puzzle and ^3He 's Minor Role

Puzzle: the equilibrium prediction gives $(D/H)_e \sim 10^{-18}$, but the **observed** ratio is

$$(D/H)_{\text{obs}} \approx 10^{-5}$$

That is a 13-order-of-magnitude mismatch – an active question in nuclear astrophysics, pointing to deuterium production *prior to* stellar formation (Big Bang nucleosynthesis). **Why ^3He plays no role in destroying d :**

$$\tau_{^3\text{He}}(d) \approx 10^4 \text{ s} \quad \text{vs.} \quad \tau_d(p) \approx 1.6 \text{ s}$$

Although $\langle\sigma v\rangle_{^3\text{He},d}$ is *seven* orders of magnitude larger than $\langle\sigma v\rangle_{pd}$, the lifetime against ^3He is four orders of magnitude *longer* – because the ^3He density is so low. $\Rightarrow$ The destruction of d is dominated entirely by $p + d \rightarrow ^3\text{He} + \gamma$.

^3He Burning and Completion of pp-I

Destruction Channels for ^3He

^3He is produced by $p + d \rightarrow ^3\text{He} + \gamma$. Possible destruction reactions:

Reaction	Q-value (MeV)
$d + ^3\text{He} \rightarrow ^5\text{Li} + \gamma$	16.38
$d + ^3\text{He} \rightarrow ^4\text{He} + p$	18.35
$^3\text{He} + ^3\text{He} \rightarrow ^6\text{Be} + \gamma$	11.50
$^3\text{He} + ^3\text{He} \rightarrow ^4\text{He} + 2p$	12.86
$^4\text{He} + ^3\text{He} \rightarrow ^7\text{Be} + \gamma$	1.58

Selecting the dominant channel: extrapolated $S(0)$ values show $d + ^3\text{He}$ and $^3\text{He} + ^3\text{He} \rightarrow ^4\text{He} + 2p$ are the key contenders. But $[d] \ll [^3\text{He}]$, so

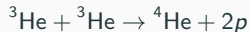
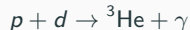
$^3\text{He} + ^3\text{He} \rightarrow ^4\text{He} + 2p$ dominates destruction of ^3He

Why $p + {}^3\text{He}$ Does Not Complete the Chain

A natural question: protons are abundant – why not



Answer: the Q -value of ${}^3\text{He}(p, \gamma){}^4\text{Li}$ is -2.5 MeV (endothermic), and ${}^4\text{Li}$ is not particle-stable. The reaction cannot proceed under stellar conditions. **Therefore the pp-I chain completes via:**



Net: $4p \rightarrow {}^4\text{He} + 2e^+ + 2\nu$ with $Q = 26.73 \text{ MeV}$. This pp-I chain operates in the cores of **first-generation stars** and in low-mass **second-generation stars** ($M \lesssim M_{\odot}$).

Equilibrium ${}^3\text{He}$ Abundance

$$\frac{d({}^3\text{He})}{dt} = r_{pd} - r_{33} = HD \langle \sigma v \rangle_{12} - \frac{({}^3\text{He})^2}{2} 2 \langle \sigma v \rangle_{33}$$

(Factor 2 in numerator: two ${}^3\text{He}$ per fusion. Factor 2 in denominator: identical-particle correction.)

Substituting $(D/H)_e = \langle \sigma v \rangle_{pp} / [2 \langle \sigma v \rangle_{pd}]$:

$$\frac{d({}^3\text{He})}{dt} = \frac{H^2 \langle \sigma v \rangle_{11}}{2} - ({}^3\text{He})^2 \langle \sigma v \rangle_{33}$$

Quasi-equilibrium ($d({}^3\text{He})/dt = 0$):

$$\boxed{\left(\frac{{}^3\text{He}}{H} \right)_e = \left(\frac{\langle \sigma v \rangle_{11}}{2 \langle \sigma v \rangle_{33}} \right)^{1/2}}$$

Lifetimes at $X_H = X_{\text{He}} = 0.5$, $\rho = 100 \text{ g/cm}^3$:

$$\tau_{{}^3\text{He}}({}^3\text{He}) \approx 10^5 \text{ y}, \quad \tau_d(\rho) \approx 1.6 \text{ s}$$

$\Rightarrow$ deuterium plays *no role* in destroying ${}^3\text{He}$ – ${}^3\text{He} + {}^3\text{He}$ dominates.

Energy Production via the $p + p$ Reaction

Energy production depends on whether ${}^3\text{He}$ has reached equilibrium. The overall rate is set by the slow $p + p$ step. **Step-by-step energy bookkeeping:**

- Fusion of 3 protons to ${}^3\text{He}$: $Q = 6.936 \text{ MeV}$, average $\langle E_\nu \rangle = 0.263 \text{ MeV}$

$$E_{\text{eff}}(3p \rightarrow {}^3\text{He}) = 6.673 \text{ MeV}$$

- Fusion of 4 protons to ${}^4\text{He}$ via ${}^3\text{He} + {}^3\text{He}$: $Q = 12.860 \text{ MeV}$

Total energy production rate: at equilibrium ($2r_{33} = r_{11}$),

$$\boxed{\varepsilon_{pp} = 13.103 r_{11} \text{ MeV cm}^{-3} \text{ s}^{-1}}$$

The overall reaction rate is entirely set by $r_{11} = r_{pp}$, so we may equivalently write $\varepsilon \approx 13 \times r_{pp}$.

The pp-II and pp-III Chains

Time for ^3He to Reach Equilibrium

Define f = fraction of ^3He abundance at time t compared to equilibrium. Let t_f = time required to reach 99% of equilibrium ($f = 0.99$). At $X_H = X_{\text{He}} = 0.5$, $\rho = 100 \text{ g/cm}^3$:

- $T_6 = 5$: $t_f \sim 10^{25} \text{ y}$ (longer than any star will ever live)
- $T_6 = 15$ (solar core): $t_f \sim 10^6 \text{ y}$ – comparable to early evolutionary timescales
- $T_6 = 30$: $t_f \sim 10^5 \text{ y}$

Consequence: the ^3He abundance *depends on the age of the star*. For the Sun, only after $\sim 10^6$ years did enough ^3He accumulate for $^3\text{He}(^3\text{He}, 2p)^4\text{He}$ to operate at its maximum rate. This is why energy-production calculations must distinguish between the pre-equilibrium and equilibrium regimes.

The Three Branches of the pp Chain

After $p + d \rightarrow {}^3\text{He} + \gamma$, the ${}^3\text{He}$ has two possible fates:

- **pp-I (86%):** ${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$
 - $Q_{\text{eff}} = 26.2 \text{ MeV}$ (2% loss to ν)
- **pp-II / pp-III (14%):** ${}^3\text{He} + {}^4\text{He} \rightarrow {}^7\text{Be} + \gamma$, then ${}^7\text{Be}$ branches:

pp-II: ${}^7\text{Be} + e^- \rightarrow {}^7\text{Li} + \nu$, ${}^7\text{Li} + p \rightarrow 2 {}^4\text{He}$

$Q_{\text{eff}} = 25.66 \text{ MeV}$ (4% ν loss) **pp-III:** ${}^7\text{Be} + p \rightarrow {}^8\text{B} + \gamma$, ${}^8\text{B} \xrightarrow{\beta^+} {}^8\text{Be}^* \rightarrow 2 {}^4\text{He}$

$Q_{\text{eff}} = 19.17 \text{ MeV}$ (28.3% ν loss – largest) In pp-II and pp-III, ${}^4\text{He}$ acts as a catalyst: an alpha enters and (effectively) two alphas come out, with 4 protons converted to ${}^4\text{He}$ overall.

Time-Evolution Equations for the pp Chains

The full set of coupled ODEs describing the abundances:

$$\begin{aligned}\frac{d(H)}{dt} &= -2\lambda_{11}\frac{H^2}{2} - \lambda_{12}HD + 2\lambda_{33}\frac{(^3\text{He})^2}{2} - \lambda_{17}H(^7\text{Be}) - \lambda_{17}^*H(^7\text{Li}) \\ \frac{d(D)}{dt} &= \lambda_{11}\frac{H^2}{2} - \lambda_{12}HD \\ \frac{d(^3\text{He})}{dt} &= \lambda_{12}HD - 2\lambda_{33}\frac{(^3\text{He})^2}{2} - \lambda_{34}(^3\text{He})(^4\text{He}) \\ \frac{d(^4\text{He})}{dt} &= \lambda_{33}\frac{(^3\text{He})^2}{2} - \lambda_{34}(^3\text{He})(^4\text{He}) + 2\lambda_{17}H(^7\text{Be}) + 2\lambda_{17}^*H(^7\text{Li}) \\ \frac{d(^7\text{Be})}{dt} &= \lambda_{34}(^3\text{He})(^4\text{He}) - \lambda_{e7}n_e(^7\text{Be}) - \lambda_{17}H(^7\text{Be}) \\ \frac{d(^7\text{Li})}{dt} &= \lambda_{e7}n_e(^7\text{Be}) - \lambda_{17}^*H(^7\text{Li})\end{aligned}$$

Each λ_{ij} is a reaction rate; minus signs $\Rightarrow$ destruction, plus signs $\Rightarrow$ production.

Net Energy Production in the Sun

Solving the abundance equations under solar conditions:

- Total energy production rate (per gram): $\varepsilon_{\text{tot}} = 5.1 \times 10^7 \text{ MeV g}^{-1} \text{ s}^{-1}$
- Observed solar luminosity: $L_{\odot} = 2.4 \times 10^{39} \text{ MeV s}^{-1}$

Mass of hydrogen actively burning:

$$m_{\odot}^{\text{burning}} = \frac{L_{\odot}}{\varepsilon_{\text{tot}}} = 4.7 \times 10^{31} \text{ g}$$

Compared to $M_{\odot} = 2 \times 10^{33} \text{ g}$:

Only $\approx 2.4\%$ of the Sun's mass is actively burning hydrogen via pp chain
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This tiny fraction is enough to power the Sun for $\sim 10^{10}$ years.

The CN Cycle

Why a Different Cycle for More Massive Stars?

Most present-day stars are **second- and third-generation** (Population I) – their gas was enriched by earlier stellar generations with elements heavier than H, He. **Selecting the next reactant:** we need an abundant nucleus with the lowest Coulomb barrier.

- Li, Be, B: very low abundance (the famous gap in the abundance curve) – excluded
- C, N: comparatively abundant; among these, ^{12}C has the lowest Coulomb barrier

⇒ The next set of reactions begins with $^{12}\text{C}(p, \gamma)^{13}\text{N}$, opening the **CN cycle**.

The CN Cycle



Key features:

- ^{12}C is consumed and regenerated $\Rightarrow$ it acts as a **catalyst**
- 4 protons are consumed; net reaction is again $4p \rightarrow ^4\text{He}$, $Q = 26.73 \text{ MeV}$
- The cycle proceeds via (p, γ) and (p, α) reactions plus β^+ decays

Rate-limiting step: governed by the *slowest* reaction = the one with the highest Coulomb barrier proceeding via EM force.

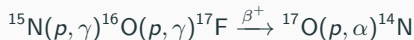


Significance: the CN cycle is **not the primary energy source in the Sun** (solar core T too low to overcome the higher Coulomb barriers), but it is essential for the **synthesis** of C, N isotopes – and it dominates energy production in more massive stars.

CNO Bi-cycle and Additional Cycles

The CNO Bi-cycle: Adding the Branch at ^{15}N

In the CN cycle we wrote $^{15}\text{N}(p, \alpha)^{12}\text{C}$. But ^{15}N can also undergo



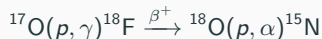
This is the **second cycle** – together with the first they form the **CNO bi-cycle**. **Relative importance of the two branches at ^{15}N :**



Ratio of $S(0)$ values: $\sim 1000 : 1$ in favor of (p, α) . $\Rightarrow$ Only about 1 in 1000 reactions follows the second branch – the bi-cycle's second loop contributes very little to the total energy production, but is essential for synthesizing heavier C/N/O isotopes.

Additional Branches: CNO3 and CNO4

The story extends further whenever ^{17}O can react via (p, γ) :



and via the fourth branch:



Branching at ^{18}O : the ratio of (p, α) vs. (p, γ) depends strongly on temperature:

- At $T_9 < 0.02$ or $T_9 > 0.7$: hydrogen burning of ^{18}O proceeds entirely via (p, α)
- In between: (p, γ) contributes

^{19}F branch: if $^{19}\text{F}(p, \alpha)^{16}\text{O}$ dominates, CNO catalysts are retained (CNO4); if $^{19}\text{F}(p, \gamma)^{20}\text{Ne}$ dominates, the cycle is broken and ^{20}Ne seeds the next cycle (NeNa). $\Rightarrow$ Four interconnected branches: CNO1, CNO2, CNO3, CNO4 – each fusing 4 protons to ^4He but synthesizing different stable isotopes.

The “Many-Cycled Monster”: Hot CNO at High T

In novae and supernovae ($T \gtrsim 10^9$ K, $M/M_\odot = 10^4$ – 10^8), the CNO cycle operates on a timescale of **seconds**. Under these conditions, β^+ -unstable nuclei (^{13}N , ^{17}F , ^{18}F , ^{19}Ne) *react with protons before they can decay*. **New pathways open up at high T :**

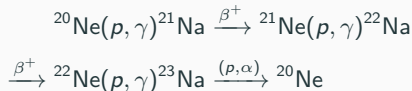
- $^{13}\text{N}(p, \gamma)^{14}\text{O} \xrightarrow{\beta^+} ^{14}\text{N}$ (bypasses $^{13}\text{N} \rightarrow ^{13}\text{C}$)
- $^{17}\text{F}(p, \gamma)^{18}\text{Ne} \xrightarrow{\beta^+} ^{18}\text{F} \rightarrow ^{15}\text{O}$
- $^{19}\text{Ne}(p, \gamma)^{20}\text{Na} \xrightarrow{\beta^+} ^{20}\text{Ne}$

This is the **hot CNO** regime – a richly interconnected network combining (p, γ) , (p, α) , and β^+ pathways. It is the gateway to heavier-element nucleosynthesis in explosive stellar environments.

NeNa and MgAl Cycles

The Neon–Sodium (NeNa) Cycle

Once ^{20}Ne is produced (from $^{19}\text{F}(p, \gamma)$), it seeds a new cycle:

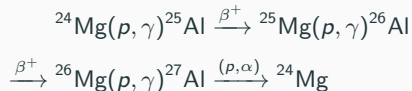


^{20}Ne acts as catalyst $\Rightarrow$ another $4p \rightarrow ^4\text{He}$ cycle. **Why this cycle matters:**

- **Not** important for energy production – the Coulomb barriers are too high
- Crucial for the **synthesis** of Ne and Na isotopes between $A = 20$ and $A = 23$
- ^{22}Ne is detected in meteorites $\Rightarrow$ direct laboratory evidence of stellar nucleosynthesis

The Magnesium–Aluminium (MgAl) Cycle

The NeNa cycle's branch $^{23}\text{Na}(p, \gamma)^{24}\text{Mg}$ initiates the MgAl cycle:



^{26}Al may also undergo (p, γ) to $^{27}\text{Si} \xrightarrow{\beta^+} ^{27}\text{Al}$, which can choose between $(p, \alpha) \rightarrow ^{24}\text{Mg}$ (cycle closes) and $(p, \gamma) \rightarrow ^{28}\text{Si}$ (cycle exits to silicon burning). **Significance:**

- ^{26}Al is found in meteorites – a famous cosmochronometer ($\tau_{1/2} \approx 7.2 \times 10^5 \text{ y}$)
- Provides isotopes between Ne and Al
- Eventual exit at ^{28}Si leads into silicon burning $\Rightarrow$ iron-peak elements
- Beyond iron, synthesis must proceed by **neutron capture** (s - and r -processes)

Nuclear Shell Model

Why We Need the Shell Model

The liquid-drop model successfully accounts for binding energies and fission, but *cannot* explain:

- The **magic numbers** (N or $Z = 2, 8, 20, 28, 50, 82, 126$) of unusually stable nuclei
- The existence and ordering of discrete nuclear excited states

Shell model advantages:

- Predicts properties of both closed and partially filled shells
- Provides the spin J and parity π of nuclear states
- Enables calculation of γ -ray transition strengths, spectroscopic factors, and weak-interaction matrix elements

Why this matters in nuclear astrophysics: every (p, γ) reaction populates a compound nucleus in an excited state. To predict its decay (gamma emission, branching ratios) we need J^π values and transition strengths – inputs that feed directly into stellar evolution codes.

Shell Model Essentials

Differences from the atomic shell model:

- No precise nuclear potential (atomic case: Coulomb is exact)
- Two species of fermions (protons and neutrons), not one (electrons)
- No heavy central force – nucleons move in their mutually generated potential

Solving the Schrödinger equation with a harmonic-oscillator or Woods–Saxon potential plus strong spin–orbit coupling gives single-particle states labelled by:

$$(n, l, j) \quad \text{with} \quad |l - 1/2| \leq j \leq l + 1/2$$

Filling rules:

- Each state of given j accommodates a maximum of $(2j + 1)$ identical nucleons (Pauli)
- Parity: $\pi = (-1)^l$ – $l = 0$ (s), 2 (d), 4 (g) $\Rightarrow$ even parity; $l = 1$ (p), 3 (f) $\Rightarrow$ odd parity
- Notation: $n l_j$ where $l = 0, 1, 2, 3, 4 \leftrightarrow s, p, d, f, g$

Filling shells reproduces the magic numbers 2, 8, 20, 28, 50, 82, 126.

Gamma Decay

Gamma Decay: Energetics

Each nuclear state has a specific excitation energy E_x , populated by capture reactions, β decay, Coulomb excitation, or scattering. The dominant de-excitation is **gamma decay**; internal conversion and internal pair production are subleading (pair production requires $E_x > 1.02$ MeV). **Energy conservation:**

$$E_\gamma = E_i - E_f - E_{\text{recoil}}$$

Recoil kinetic energy is typically $\sim$ eV $\Rightarrow$ negligible, so

$$E_\gamma \approx E_i - E_f$$

X-rays vs. gamma rays: same electromagnetic radiation, different origin:

- X-rays: atomic electron rearrangements
- Gamma rays: nuclear transitions

Lab example: $^{137}\text{Cs} \xrightarrow{\beta^-} ^{137}\text{Ba}^* \text{ (excited)} \xrightarrow{\gamma} ^{137}\text{Ba} \text{ (ground)}$ – gamma emission is essentially always preceded by β decay.

Gamma Decay: Angular Momentum and Multipolarity

The photon must conserve both angular momentum and parity. Define the multipolarity:

$$|I_f - I_i| \leq I \leq I_f + I_i \quad (\text{vector coupling})$$

The change in intrinsic angular momentum is $\Delta\hbar = (I_f - I_i)\hbar$. The emitted photon's multipolarity is I , with radiation pattern 2^I -pole:

$$I = 1 \text{ (dipole)}, \quad I = 2 \text{ (quadrupole)}, \quad I = 3 \text{ (octupole)}, \dots$$

Electric vs. magnetic character (governed by parity change):

$$\Delta\pi = (-1)^I \Rightarrow \text{electric multipole} \quad \Delta\pi = (-1)^{I+1} \Rightarrow \text{magnetic multipole}$$

Type	Nature	I	$\Delta\pi$
E_1	Electric dipole	1	Yes
M_1	Magnetic dipole	1	No
E_2	Electric quadrupole	2	No
M_2	Magnetic quadrupole	2	Yes
E_3	Electric octupole	3	Yes
M_3	Magnetic octupole	3	No

Selection-Rule Example: ^{23}Na

Consider a ^{23}Na transition from $\frac{7}{2}^{+}$ to $\frac{3}{2}^{+}$. **Step 1 – Parity change:** both states have positive parity $\Rightarrow \Delta\pi = \text{No}$. **Step 2 – Allowed l values:**

$$\left| \frac{7}{2} - \frac{3}{2} \right| \leq l \leq \frac{7}{2} + \frac{3}{2} \Rightarrow 2 \leq l \leq 5$$

Step 3 – Classify each allowed l :

l	$\Delta\pi$ required	Type
2	No	E_2
3	No	M_3
4	No	E_4
5	No	M_5

Lowest multipolarities dominate in practice $\Rightarrow$ the transition is predominantly E_2 , with progressively smaller M_3 , E_4 , M_5 contributions.

Summary

Summary (1/2): pp Chains and Burning Stages

- Deuterium burning is dominated by $p + d \rightarrow {}^3\text{He} + \gamma$; quasi-equilibrium gives $(D/H)_e \sim 10^{-18}$, in stark contrast to the observed $\sim 10^{-5}$ – a Big-Bang signature
- ${}^3\text{He}$ burning completes pp-I via ${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$; protons cannot complete the chain because ${}^4\text{Li}$ is not particle-stable
- Three branches of the pp chain split at ${}^3\text{He}$:
 - pp-I (86%, $Q_{\text{eff}} = 26.2$ MeV)
 - pp-II (14% combined, via ${}^7\text{Be}$ electron capture, $Q_{\text{eff}} = 25.66$ MeV)
 - pp-III (small, via ${}^8\text{B}$, $Q_{\text{eff}} = 19.17$ MeV – largest ν loss)
- Net energy production: $\varepsilon = 13.1 r_{11} \text{ MeV cm}^{-3} \text{ s}^{-1}$, set entirely by the slow $p + p$ rate
- Only about 2.4% of the Sun's mass is involved in active pp burning

Summary (2/2): Cycles, Shell Model, Gamma Decay

- CN cycle (rate-limited by $^{14}\text{N}(p, \gamma)^{15}\text{O}$) is the dominant H-burning channel above $T_6 \approx 20$, and is critical for C/N isotope synthesis
- CNO bi-cycle adds the $^{15}\text{N}(p, \gamma)^{16}\text{O}$ branch (1000:1 against (p, α)); CNO3 and CNO4 extend to O, F, and Ne isotopes
- Hot CNO (novae, supernovae): β^+ -unstable nuclei react with protons before decaying $\Rightarrow$ richly interconnected “many-cycled monster”
- NeNa and MgAl cycles synthesize elements between ^{20}Ne and ^{27}Al – not significant for energy, but produce ^{22}Ne and ^{26}Al found in meteorites; ^{28}Si seeds the next burning stage
- Shell model: harmonic / Woods–Saxon potential with spin–orbit coupling gives (n, l, j) single-particle states; explains magic numbers and supplies J^π , transition strengths
- Gamma decay: $E_\gamma \approx E_i - E_f$; multipolarity determined by $|I_f - I_i| \leq l \leq I_f + I_i$; electric vs. magnetic character set by parity change via $(-1)^l$ vs. $(-1)^{l+1}$